

COSC 594

Policy Optimization (I)

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Announcement

Check Canvas in your myutk account.

Outline

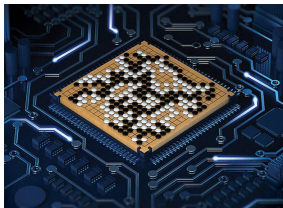
- POLICY OPTIMIZATION
- POLICY GRADIENT METHODS

Outline

■ POLICY OPTIMIZATION

Policy optimization in RL

Go



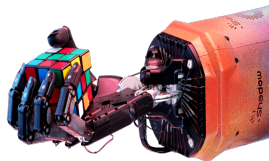
AlphaZero, Silver et al., '17

Video game



OpenAI Five, '18

Robotics



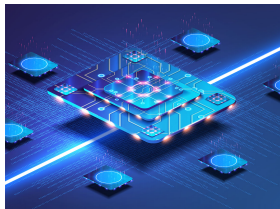
Rubik's cube, OpenAI, '19

Automated trading



Crypto trading bots, '20

Chip design

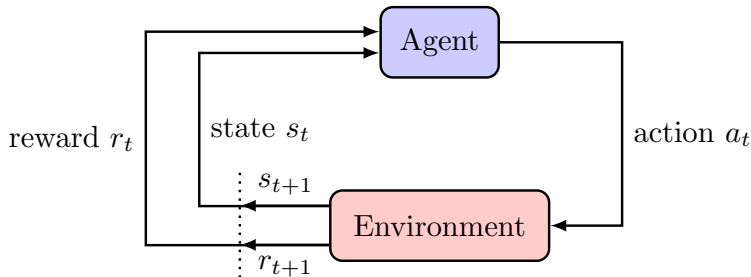


AI chip, Azalia et al., Google, '21

etc.

RL Framework

■ MARKOV DECISION PROCESSES (MDPs)



Goal $\underset{\pi}{\text{maximize}} \quad \mathbb{E}_{s_0} [V^{\pi}(s_0)]$	$\Leftrightarrow$	Direct policy search $\pi^+ \leftarrow \pi + \nabla_{\pi} V^{\pi}$
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π – a policy that maps states to actions

$V^{\pi}(s_0) := \mathbb{E} [\text{sum of all rewards received}]$

Why is it widely used?

■ OPTIMIZATION

Optimization algorithms are extendable.

■ HANDLES LARGE STATE AND ACTION SPACES

Neural networks are used to represent policies.

■ MAKES USE OF POLICY SIMULATORS

Policies are easily rolled out to estimate their values.

■ DIRECTLY OPTIMIZES OBJECTIVES OF INTEREST

Value functions of interest are tracked directly.

Policy optimization

Policy: $\pi : S \text{ (states)} \rightarrow \Delta_A \text{ (action distribution)}$

Value function: $V^\pi(\rho) := \mathbb{E} \left[\sum_{t=0}^{\infty} \gamma^t r(s_t, a_t) \mid s_0 \sim \rho \right]$

$P(\cdot \mid s, a) : S \times A \rightarrow \Delta_A$ – transition kernel

$s_0, a_0, r_0, s_1, a_1, \dots$ – a trajectory τ generated by a policy π

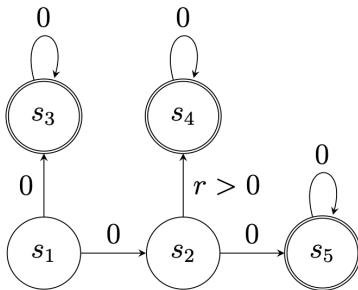
■ NONCONVEX POLICY OPTIMIZATION

$$\underset{\pi}{\text{maximize}} \quad V^\pi(\rho)$$

★ Search the decision in [a function space](#)

Nonconvex example

■ A DETERMINISTIC MDP



s_3, s_4, s_5 – terminal states

a_1 (up/above), a_2 (right) – actions

$V^\pi(s_1) = \pi(a_2 | s_1)\pi(a_1 | s_2)r$ – value function in s_1

★ No global optimality for maximizing nonconcave functions

Policy parametrization

Policy: $\pi_{\theta}(\cdot | s): S \rightarrow \Delta_A$

Parameter: $\theta \in \mathbb{R}^d$

■ TABULAR POLICY

$$\pi_{\theta}(a | s) = \theta_{s,a} \text{ w/ parameter } \theta \in \mathbb{R}^{|S||A|}$$

■ SOFTMAX POLICY

$$\pi_{\theta}(a | s) = \frac{e^{\theta_{s,a}}}{\sum_{a'} e^{\theta_{s,a'}}} \text{ w/ parameter } \theta \in \mathbb{R}^{|S||A|}$$

■ LOG-LINEAR POLICY

$$\pi_{\theta}(a | s) = \frac{e^{\theta^{\top} \phi_{s,a}}}{\sum_{a'} e^{\theta^{\top} \phi_{s,a'}}} \text{ w/ feature } \phi_{s,a} \in \mathbb{R}^d$$

Outline

- POLICY OPTIMIZATION
- POLICY GRADIENT METHODS

Policy gradient methods

Policy: $\pi_{\theta}(\cdot | s): S \rightarrow \Delta_A$

Parameter: $\theta \in \mathbb{R}^d$

■ NONCONVEX PARAMETER OPTIMIZATION

$$\underset{\theta}{\text{maximize}} \quad V^{\pi_{\theta}}(\rho)$$

■ GRADIENT ASCENT

$$\theta^+ \leftarrow \theta + \eta \nabla_{\theta} V^{\pi_{\theta}}(\rho)$$

★ Search the decision in [a parameter space](#)

Policy gradient: REINFORCE version

$$\nabla_{\theta} V^{\pi_{\theta}}(\rho) = \mathbb{E}_{\tau} \left[\left(\sum_{t=0}^{\infty} \gamma^t r_t \right) \left(\sum_{t=0}^{\infty} \nabla_{\theta} \log \pi_{\theta}(a_t | s_t) \right) \right]$$

■ LINEAR FORM OF VALUE FUNCTION

$$V^{\pi_{\theta}}(\rho) = \sum_{\tau} \Pr(\tau | \pi_{\theta}) R(\tau)$$

$R(\tau) := \sum_{t=0}^{\infty} \gamma^t r_t$ – total rewards over a trajectory τ

$\Pr(\tau | \pi_{\theta}) := \rho(s_0) \pi_{\theta}(a_0 | s_0) P(s_1 | s_0, a_0) \cdots$ – probability of a trajectory τ

■ LOG-DERIVATIVE TRICK

$$\begin{aligned}\nabla_{\theta} V^{\pi_{\theta}}(\rho) &= \sum_{\tau} \nabla_{\theta} \Pr(\tau \mid \pi_{\theta}) R(\tau) \\ &= \sum_{\tau} \Pr(\tau \mid \pi_{\theta}) \nabla_{\theta} \log \Pr(\tau \mid \pi_{\theta}) R(\tau) \\ &= \sum_{\tau} \Pr(\tau \mid \pi_{\theta}) R(\tau) \sum_{t=0}^{\infty} \nabla_{\theta} \log \pi_{\theta}(a_t \mid s_t)\end{aligned}$$

- ★ Unbiased gradient estimation with samples
- ★ Expensive policy rollouts

Policy gradient: state-action value version

$$\nabla_{\theta} V^{\pi_{\theta}}(\rho) = \mathbb{E}_{\tau} \left[\sum_{t=0}^{\infty} \gamma^t Q^{\pi_{\theta}}(s_t, a_t) \nabla_{\theta} \log \pi_{\theta}(a_t | s_t) \right]$$

■ STATE-ACTION VALUE FUNCTION

$$Q^{\pi}(s, a) := \mathbb{E} \left[\sum_{t=0}^{\infty} \gamma^t r(s_t, a_t) \mid s_0 = s, a_0 = a \right]$$

$Q^{\pi}(s, a) = r(s, a) + \gamma \sum_{s'} P(s' | s, a) V^{\pi}(s')$ – state-action value function

$V^{\pi}(s) = \mathbb{E}_{a \sim \pi(\cdot | s)} [Q^{\pi}(s, a)]$ – state value function

■ PRODUCT DERIVATIVE

$$\begin{aligned}\nabla_{\theta} V^{\pi_{\theta}}(s_0) &= \nabla_{\theta} \sum_{a_0} \pi_{\theta}(a_0 | s_0) Q^{\pi_{\theta}}(s_0, a_0) \\&= \sum_{a_0} \nabla_{\theta} \pi_{\theta}(a_0 | s_0) Q^{\pi_{\theta}}(s_0, a_0) + \sum_{a_0} \pi_{\theta}(a_0 | s_0) \nabla_{\theta} Q^{\pi_{\theta}}(s_0, a_0) \\&= \sum_{a_0} \pi_{\theta}(a_0 | s_0) \nabla_{\theta} \log \pi_{\theta}(a_0 | s_0) Q^{\pi_{\theta}}(s_0, a_0) \\&\quad + \sum_{a_0} \pi_{\theta}(a_0 | s_0) \nabla_{\theta} \left(r(s_0, a_0) + \gamma \sum_{s_1} P(s_1 | s_0, a_0) V^{\pi_{\theta}}(s_1) \right) \\&= \mathbb{E}_{\tau} [\nabla_{\theta} \log \pi_{\theta}(a_0 | s_0) Q^{\pi_{\theta}}(s_0, a_0)] + \gamma \mathbb{E}_{\tau} [\nabla_{\theta} V^{\pi_{\theta}}(s_1)]\end{aligned}$$

★ Recursive policy gradient

■ GRADIENT RECURSION

$$\begin{aligned}\nabla_{\theta} V^{\pi_{\theta}}(\rho) &= \mathbb{E}_{\tau} [\nabla_{\theta} \log \pi_{\theta}(a_0 | s_0) Q^{\pi_{\theta}}(s_0, a_0)] + \gamma \mathbb{E}_{\tau} [\nabla_{\theta} V^{\pi_{\theta}}(s_1)] \\&= \mathbb{E}_{\tau} [\nabla_{\theta} \log \pi_{\theta}(a_0 | s_0) Q^{\pi_{\theta}}(s_0, a_0)] + \gamma \mathbb{E}_{\tau} [\nabla_{\theta} \log \pi_{\theta}(a_1 | s_1) Q^{\pi_{\theta}}(s_1, a_1)] \\&\quad + \gamma^2 \mathbb{E}_{\tau} [\nabla_{\theta} V^{\pi_{\theta}}(s_2)] \\&= \dots \\&= \mathbb{E}_{\tau} \left[\sum_{t=0}^{\infty} \gamma^t Q^{\pi_{\theta}}(s_t, a_t) \nabla_{\theta} \log \pi_{\theta}(a_t | s_t) \right]\end{aligned}$$

★ Unbiased gradient estimation with state-action values

$$\nabla_{\theta} V^{\pi_{\theta}}(\rho) = \frac{1}{1-\gamma} \mathbb{E}_{s \sim d_{\rho}^{\pi_{\theta}}(\cdot), a \sim \pi_{\theta}(\cdot | s)} [Q^{\pi_{\theta}}(s, a) \nabla_{\theta} \log \pi_{\theta}(a | s)]$$

Policy gradient: advantage version

$$\nabla_{\theta} V^{\pi_{\theta}}(\rho) = \frac{1}{1-\gamma} \mathbb{E}_{s \sim d_{\rho}^{\pi_{\theta}}(\cdot), a \sim \pi_{\theta}(\cdot | s)} [A^{\pi_{\theta}}(s, a) \nabla_{\theta} \log \pi_{\theta}(a | s)]$$

$$\sum_a \pi_{\theta}(a | s) = 1 \quad - \quad \text{valid policy}$$

■ ADVANTAGE FUNCTION

$$A^{\pi}(s, a) \quad := \quad Q^{\pi}(s, a) - V^{\pi}(s)$$

$$\mathbb{E}_{a \sim \pi(\cdot | s)} [A^{\pi}(s, a)] = 0 \quad - \quad \text{zero advantage}$$

Summary

■ POLICY OPTIMIZATION

Nonconvex optimization problem

■ POLICY GRADIENT METHODS

REINFORCE, state-action value version, advantage version, unbiased gradient estimation